

PES UNIVERSITY, BENGALURU

Department of Computer Science and Engineering
B.Tech. Computer Science and Engineering | V Semester
UE24CS341A — SOFTWARE ENGINEERING (JACKFRUIT)

SOFTWARE ARCHITECTURE

ChargeSlot — EV Charging Station Finder with Slot Dashboard and Auto-Booking System

Team Number: 19 | Project ID: 37 | Team Name: NoteWare

Member	SRN	Owned functionality
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Document version: 0.1 (proposed / pre-implementation) | 1 October 2026

Source: ChargeSlot SRS v1.0 (5 September 2026) and Jackfruit SE Guidelines 2026.

Revision History

Name	Date	Reason for Changes	Version
Team NoteWare	1 October 2026	Initial planning draft; implementation and validation pending	0.1

1. Introduction and Architectural Goals

Design status: proposed high-level architecture, not an implemented system. Source of truth: ChargeSlot SRS v1.0. Goals: safe booking concurrency, role-scoped privacy, simulator portability, traceable billing, responsive live status, independent work by four feature owners.

2. Constraints and Consequences

SRS constraint	Architectural consequence
React Native TypeScript + FastAPI Python + PostgreSQL/PostGIS fixed	One mobile client, typed REST contracts, geospatial queries in PostGIS; avoid parallel stacks.
Redis slot holds and live connector cache	Short-lived status in Redis; PostgreSQL is durable booking source of truth; no Redis-only confirmed booking.
No real chargers; OCPP 1.6J simulator	Gateway adapter isolates protocol; physical charger behavior remains unverified.
Razorpay test mode; no card storage	Hosted checkout; signed webhooks, server verification and idempotency required.
Google Maps quotas	On-demand route calls, bounded TTL caching and fallback; no per-refresh Directions call.
JWT and server-side RBAC	All object reads/writes filtered by principal, ownership and operator station scope.
Jenkins, Docker, pytest, SonarQube and reviewed PRs	Build/test/security gates become integration prerequisites, not optional afterthoughts.

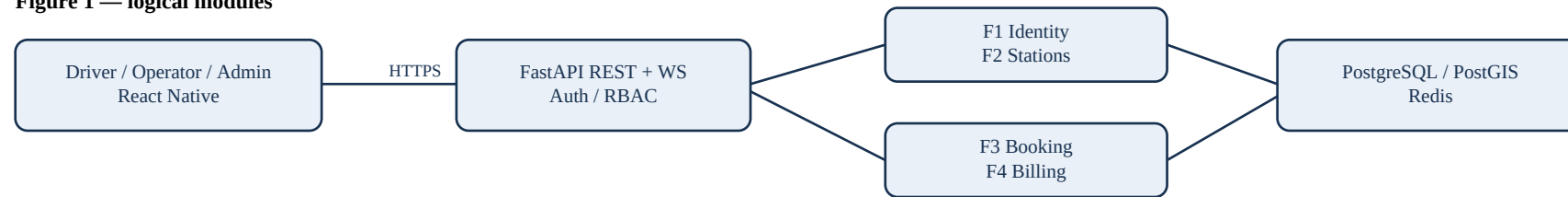
3. Architectural Style and Alternatives

Decision: modular monolith with explicit domain services, layered adapters and one FastAPI deployment, plus a React Native mobile client. PostgreSQL transactions provide a single consistency boundary for bookings and settlement; each member owns a bounded module and public contract.

Alternative	Why rejected now	Cost of chosen style
Microservices	Four student owners cannot justify service discovery, distributed transactions, separate deployment pipelines and cross-service debugging for a prototype.	Shared deploy cadence; failure isolation is weaker.
Single-layer MVC	Controllers directly mixing gateway calls, booking rules and persistence would make RBAC and transaction invariants hard to test.	More interface/DTO boilerplate.
Event-driven core	Asynchronous booking/payment commands risk eventual consistency and duplicate side effects in the critical reservation path.	Background notifications and analytics still need queue/retry design.

4. Logical View

Figure 1 — logical modules



5. Component View

Figure 2 — component boundaries



6. Deployment View

Figure 3 — proposed deployment topology



Jenkins builds and tests container images; GitHub PR review is mandatory. Secrets injected from CI/runtime configuration, not checked into repository. Cloud vendor is deliberately unspecified by the SRS.

7. Process View — Booking and Settlement

Figure 4 — sequence of major responsibilities



Sequence contract: (1) JWT and vehicle ownership checked; (2) availability and compatible connector validated; (3) atomic slot hold with expiry; (4) hosted payment or reservation fee as specified; (5) verify signed webhook and server-side gateway status; (6) idempotently confirm; (7) check-in and simulated charging session; (8) meter readings produce immutable tariff-snapshot invoice; (9) settlement and analytics update. Failure returns controlled state, not a second booking or duplicate charge.

8. Data View

Figure 5 — conceptual ER relationships



Persistent SRS entities: User, Vehicle, Station, Charger, Connector, Slot, Booking, Charging Session, Payment and analytics aggregates. Add invoice, tariff snapshot, audit and payment idempotency records as architectural design entities; verify physical schema in the subsequent SDD. Foreign keys enforce owner and station association. Booking state machine: Held → Confirmed → Active → Completed, with Cancelled, No-Show, Expired and Reassigned transitions only as defined in SRS.

9. Module Interaction Rules

Only owning module writes its aggregate. F1 owns principal/vehicle access; F2 owns stations, connector status and slot availability; F3 owns hold/booking/session lifecycle; F4 owns payment, invoice and aggregates. Modules call versioned service interfaces, never another module's ORM tables directly. Booking uses PostgreSQL row lock or exclusion constraint for overlaps; event/outbox for notifications and analytics must be idempotent. Shared DTO schema and error envelope are reviewed before independent implementation. Each owner provides contract tests and fake adapters; PRs changing a contract require consumer sign-off.

10. REST API Surface — Proposed Contract

Routes below are design proposals derived from SRS features, not pre-existing endpoints. Each endpoint maps to the specific requirement it implements. WebSocket and OCPP adapter paths are separately marked.

10.1 User Accounts, Vehicle Profile and Access Control

Method	Proposed route / operation	Requirement ID	Responsibility
POST	/api/v1/accounts/register-account	REQ-F1-01	Register account
GET	/api/v1/accounts/password-policy	REQ-F1-02	Password policy
POST	/api/v1/accounts/verify-contact-details	REQ-F1-03	Verify contact details
POST	/api/v1/accounts/login-and-tokens	REQ-F1-04	Login and tokens
POST	/api/v1/accounts/login-lockout	REQ-F1-05	Login lockout
POST	/api/v1/accounts/password-reset	REQ-F1-06	Password reset
POST	/api/v1/accounts/logout	REQ-F1-07	Logout
GET	/api/v1/accounts/reject-duplicate-account	REQ-F1-08	Reject duplicate account
POST	/api/v1/accounts/generic-login-error	REQ-F1-09	Generic login error
PATCH	/api/v1/accounts/assign-one-role	REQ-F1-10	Assign one role
GET	/api/v1/accounts/server-side-rbac	REQ-F1-11	Server-side RBAC
GET	/api/v1/accounts/operator-data-scope	REQ-F1-12	Operator data scope
PATCH	/api/v1/accounts/admin-user-management	REQ-F1-13	Admin user management
GET	/api/v1/accounts/audit-user-actions	REQ-F1-14	Audit user actions
GET	/api/v1/accounts/operator-onboarding	REQ-F1-15	Operator onboarding
PATCH	/api/v1/accounts/edit-profile-and-preferences	REQ-F1-16	Edit profile and preferences
GET	/api/v1/accounts/profile-validation	REQ-F1-17	Profile validation
GET	/api/v1/accounts/manual-location-fallback	REQ-F1-18	Manual location fallback
GET	/api/v1/accounts/location-retention-limit	REQ-F1-19	Location retention limit
GET	/api/v1/accounts/account-deletion	REQ-F1-20	Account deletion
GET/POST/PATCH/DELETE	/api/v1/accounts/vehicle-crud	REQ-F1-21	Vehicle CRUD
GET	/api/v1/accounts/vehicle-limit-and-default	REQ-F1-22	Vehicle limit and default
POST	/api/v1/accounts/preset-ev-models	REQ-F1-23	Preset EV models
GET	/api/v1/accounts/vehicle-value-validation	REQ-F1-24	Vehicle value validation
POST	/api/v1/accounts/protect-booked-vehicle	REQ-F1-25	Protect booked vehicle
GET	/api/v1/accounts/driver-sees-own-data-only	REQ-F1-26	Driver sees own data only

10.2 Station Discovery, Navigation and Real-Time Slot Dashboard

Method	Proposed route / operation	Requirement ID	Responsibility
GET/POST/PATCH/DELETE	/api/v1/stations/station-crud	REQ-F2-01	Station CRUD
GET/POST/PATCH/DELETE	/api/v1/stations/charger-and-connector-crud	REQ-F2-02	Charger and connector CRUD
GET	/api/v1/stations/station-approval-gate	REQ-F2-03	Station approval gate
GET	/api/v1/stations/station-validation	REQ-F2-04	Station validation
PATCH	/api/v1/stations/out-of-service-marking	REQ-F2-05	Out-of-service marking
GET	/api/v1/stations/maintenance-windows	REQ-F2-06	Maintenance windows
GET	/api/v1/stations/nearby-station-search	REQ-F2-07	Nearby station search
PATCH	/api/v1/stations/availability-markers	REQ-F2-08	Availability markers
GET	/api/v1/stations/filter-stations	REQ-F2-09	Filter stations
GET	/api/v1/stations/sort-stations	REQ-F2-10	Sort stations
GET	/api/v1/stations/text-search	REQ-F2-11	Text search
GET	/api/v1/stations/pagination-and-clustering	REQ-F2-12	Pagination and clustering
GET	/api/v1/stations/station-detail	REQ-F2-13	Station detail
GET	/api/v1/stations/favourite-stations	REQ-F2-14	Favourite stations
POST	/api/v1/stations/rate-a-station	REQ-F2-15	Rate a station
POST	/api/v1/stations/report-faulty-charger	REQ-F2-16	Report faulty charger
OCPP	/gateway/ocpp/1.6j	REQ-F2-17	Ingest connector status
GET	/api/v1/stations/live-status-display	REQ-F2-18	Live status display
WS	/ws/stations/status	REQ-F2-19	Live status push
GET	/api/v1/stations/stale-status-handling	REQ-F2-20	Stale status handling
GET	/api/v1/stations/reserved-status	REQ-F2-21	Reserved status
GET	/api/v1/stations/slot-generation	REQ-F2-22	Slot generation
GET	/api/v1/stations/slot-timeline	REQ-F2-23	Slot timeline
GET	/api/v1/stations/estimated-wait-time	REQ-F2-24	Estimated wait time
GET	/api/v1/stations/route-preview	REQ-F2-25	Route preview
GET	/api/v1/stations/navigation-hand-off	REQ-F2-26	Navigation hand-off
GET	/api/v1/stations/eta-versus-slot-check	REQ-F2-27	ETA versus slot check
GET	/api/v1/stations/route-cache-and-fallback	REQ-F2-28	Route cache and fallback

10.3 Battery-Aware Booking and Auto-Booking

Method	Proposed route / operation	Requirement ID	Responsibility
GET	/api/v1/bookings/battery-level-input	REQ-F3-01	Battery level input
GET	/api/v1/bookings/battery-input-validation	REQ-F3-02	Battery input validation
GET	/api/v1/bookings/usable-range-calculation	REQ-F3-03	Usable range calculation
GET	/api/v1/bookings/reachability-filter	REQ-F3-04	Reachability filter
GET	/api/v1/bookings/connector-compatibility	REQ-F3-05	Connector compatibility
GET	/api/v1/bookings/energy-estimate	REQ-F3-06	Energy estimate
GET	/api/v1/bookings/charging-time-estimate	REQ-F3-07	Charging time estimate
GET	/api/v1/bookings/slot-length-rule	REQ-F3-08	Slot length rule
GET	/api/v1/bookings/cost-estimate	REQ-F3-09	Cost estimate
GET	/api/v1/bookings/recommended-slots	REQ-F3-10	Recommended slots
GET	/api/v1/bookings/manual-slot-choice	REQ-F3-11	Manual slot choice
GET	/api/v1/bookings/slot-hold	REQ-F3-12	Slot hold
POST	/api/v1/bookings/no-double-booking	REQ-F3-13	No double booking
GET	/api/v1/bookings/driver-overlap-rule	REQ-F3-14	Driver overlap rule
POST	/api/v1/bookings/confirm-booking	REQ-F3-15	Confirm booking
POST	/api/v1/bookings/auto-book-action	REQ-F3-16	Auto-book action
GET	/api/v1/bookings/weighted-slot-scoring	REQ-F3-17	Weighted slot scoring
GET	/api/v1/bookings/explain-the-choice	REQ-F3-18	Explain the choice
POST	/api/v1/bookings/confirmation-modes	REQ-F3-19	Confirmation modes
GET	/api/v1/bookings/low-battery-trigger	REQ-F3-20	Low-battery trigger
GET	/api/v1/bookings/no-slot-handling	REQ-F3-21	No-slot handling
POST	/api/v1/bookings/booking-state-machine	REQ-F3-22	Booking state machine
POST	/api/v1/bookings/modify-booking	REQ-F3-23	Modify booking
POST	/api/v1/bookings/cancel-booking	REQ-F3-24	Cancel booking
GET	/api/v1/bookings/no-show-handling	REQ-F3-25	No-show handling
POST	/api/v1/bookings/auto-rebooking	REQ-F3-26	Auto-rebooking
POST	/api/v1/bookings/check-in	REQ-F3-27	Check-in
POST	/api/v1/bookings/remote-start	REQ-F3-28	Remote start

Method	Proposed route / operation	Requirement ID	Responsibility
GET	/api/v1/bookings/live-session-monitor	REQ-F3-29	Live session monitor
POST	/api/v1/bookings/stop-session	REQ-F3-30	Stop session
POST	/api/v1/bookings/start-failure-handling	REQ-F3-31	Start failure handling
GET	/api/v1/bookings/idle-fee	REQ-F3-32	Idle fee
POST	/api/v1/bookings/booking-notifications	REQ-F3-33	Booking notifications

10.4 Payments, Billing and Usage Analytics

Method	Proposed route / operation	Requirement ID	Responsibility
GET	/api/v1/billing/payment-methods	REQ-F4-01	Payment methods
GET	/api/v1/billing/tokenised-payment-methods	REQ-F4-02	Tokenised payment methods
POST	/api/v1/billing/remove-payment-method	REQ-F4-03	Remove payment method
GET	/api/v1/billing/reservation-fee	REQ-F4-04	Reservation fee
GET	/api/v1/billing/final-bill-computation	REQ-F4-05	Final bill computation
GET	/api/v1/billing/tariff-snapshot	REQ-F4-06	Tariff snapshot
POST	/api/v1/billing/settle-balance	REQ-F4-07	Settle balance
POST	/api/v1/billing/refunds	REQ-F4-08	Refunds
GET	/api/v1/billing/payment-records	REQ-F4-09	Payment records
GET	/api/v1/billing/idempotent-retry	REQ-F4-10	Idempotent retry
GET	/api/v1/billing/webhook-verification	REQ-F4-11	Webhook verification
GET	/api/v1/billing/server-side-verification	REQ-F4-12	Server-side verification
POST	/api/v1/billing/daily-reconciliation	REQ-F4-13	Daily reconciliation
GET	/api/v1/billing/invoice-generation	REQ-F4-14	Invoice generation
GET	/api/v1/billing/payment-history	REQ-F4-15	Payment history
PATCH	/api/v1/billing/tariff-management	REQ-F4-16	Tariff management
GET	/api/v1/billing/operator-earnings	REQ-F4-17	Operator earnings
GET	/api/v1/billing/driver-usage-summary	REQ-F4-18	Driver usage summary
GET	/api/v1/billing/driver-trend-charts	REQ-F4-19	Driver trend charts
GET	/api/v1/billing/top-stations-and-co2	REQ-F4-20	Top stations and CO2
GET	/api/v1/billing/operator-analytics	REQ-F4-21	Operator analytics

Method	Proposed route / operation	Requirement ID	Responsibility
GET	/api/v1/billing/platform-analytics	REQ-F4-22	Platform analytics
GET	/api/v1/billing/date-range-filter	REQ-F4-23	Date range filter
GET	/api/v1/billing/csv-export	REQ-F4-24	CSV export
GET	/api/v1/billing/role-scoped-analytics	REQ-F4-25	Role-scoped analytics
GET	/api/v1/billing/analytics-refresh	REQ-F4-26	Analytics refresh

11. Internal Module Contracts

Provider → Consumer	Proposed interface	Input → output / failure
F1 → F2/F3/F4	authorize(principal, action, resource_id)	JWT principal + object → allow/deny; 401/403/404
F2 → F3	get_available_slots(station_id, connector_type, interval)	Station/connector + window → free slots; stale/maintenance excluded
F1 → F3	get_owned_vehicle(principal, vehicle_id)	Vehicle attributes or 404; never cross-owner data
F3 → F4	get_completed_session(session_id)	Energy, timestamps, tariff snapshot, booking ID; immutable settlement inputs
F4 → F3	verify_reservation_payment(booking_id, gateway_ref)	Verified amount and idempotent payment state or failure
OCPP adapter → F2/F3	ingest_status(connector_id, state, observed_at)	Versioned status + timestamp; reject replay/out-of-order
F3/F4 → notifications	enqueue_notice(event_id, recipient, payload)	At-least-once delivery with deduplication by event ID

12. Cross-Cutting Concerns

Security: JWT expiry and refresh rotation; Argon2/bcrypt passwords; RBAC + object-level ownership; signed webhook validation; HTTPS; no card PAN storage; input validation and rate limits. Errors: versioned JSON envelope {code, message, request_id}, safe 4xx messages and no stack traces. Caching: Redis TTL for status/slot holds, cached Maps routes with staleness indication, invalidate on connector events. Logging: structured request ID, principal pseudonymous ID, action, outcome and timestamp; redact tokens, payment data and location. Configuration: environment-backed secrets, separate dev/test/prod values, CI secret store. Resilience: bounded retries, circuit breaker for maps/gateway, idempotency keys, durable outbox for asynchronous side effects.

13. Architecture Decision Records

ADR	Context	Decision	Consequences
ADR-01 Modular monolith	Four parallel owners, short academic schedule	Single FastAPI deployment with strict module interfaces	Simpler transactional integrity; shared deployment/fault domain
ADR-02 Durable booking state	Redis hold expiry and concurrent reservations	PostgreSQL atomic booking invariants; Redis only transient	Less overselling risk; database contention and transaction design needed
ADR-03 Simulator-first	No physical charger available	OCPP 1.6J adapter and simulator	Repeatable demos; hardware reliability not verified

ADR	Context	Decision	Consequences
ADR-04 Hosted payment	Card data protection, sandbox-only	Razorpay hosted checkout + signed webhook + server verification	Reduced PCI scope; gateway availability and callback races
ADR-05 Live status channel	Real-time dashboard requirement	WebSocket push plus poll fallback and staleness TTL	Better latency; socket lifecycle and state-state complexity
ADR-06 Rule-based ranking	SRS prohibits trained ML in Phase 1	Replaceable weighted scoring interface	Explainable deterministic output; less adaptive than learned model

14. Requirement-to-Architecture Traceability

Requirement ID	SRS description	SRS architecture reference	Owning module / contract
REQ-F1-01	Register account	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-02	Password policy	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-03	Verify contact details	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-04	Login and tokens	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-05	Login logout	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-06	Password reset	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-07	Logout	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-08	Reject duplicate account	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-09	Generic login error	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-10	Assign one role	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-11	Server-side RBAC	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-12	Operator data scope	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-13	Admin user management	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-14	Audit user actions	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-15	Operator onboarding	HLD §Auth	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-16	Edit profile and preferences	HLD §Profile	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-17	Profile validation	HLD §Profile	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-18	Manual location fallback	HLD §Profile	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-19	Location retention limit	HLD §Profile	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-20	Account deletion	HLD §Profile	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-21	Vehicle CRUD	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control

Requirement ID	SRS description	SRS architecture reference	Owning module / contract
REQ-F1-22	Vehicle limit and default	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-23	Preset EV models	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-24	Vehicle value validation	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-25	Protect booked vehicle	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control
REQ-F1-26	Driver sees own data only	HLD §Vehicles	F1: User Accounts, Vehicle Profile and Access Control
REQ-F2-01	Station CRUD	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-02	Charger and connector CRUD	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-03	Station approval gate	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-04	Station validation	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-05	Out-of-service marking	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-06	Maintenance windows	HLD §Stations	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-07	Nearby station search	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-08	Availability markers	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-09	Filter stations	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-10	Sort stations	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-11	Text search	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-12	Pagination and clustering	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-13	Station detail	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-14	Favourite stations	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-15	Rate a station	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-16	Report faulty charger	HLD §Discovery	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-17	Ingest connector status	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-18	Live status display	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-19	Live status push	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-20	Stale status handling	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-21	Reserved status	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-22	Slot generation	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-23	Slot timeline	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard

Requirement ID	SRS description	SRS architecture reference	Owning module / contract
REQ-F2-24	Estimated wait time	HLD §Dashboard	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-25	Route preview	HLD §Navigation	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-26	Navigation hand-off	HLD §Navigation	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-27	ETA versus slot check	HLD §Navigation	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F2-28	Route cache and fallback	HLD §Navigation	F2: Station Discovery, Navigation and Real-Time Slot Dashboard
REQ-F3-01	Battery level input	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-02	Battery input validation	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-03	Usable range calculation	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-04	Reachability filter	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-05	Connector compatibility	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-06	Energy estimate	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-07	Charging time estimate	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-08	Slot length rule	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-09	Cost estimate	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-10	Recommended slots	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-11	Manual slot choice	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-12	Slot hold	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-13	No double booking	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-14	Driver overlap rule	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-15	Confirm booking	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-16	Auto-book action	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-17	Weighted slot scoring	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-18	Explain the choice	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-19	Confirmation modes	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-20	Low-battery trigger	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-21	No-slot handling	HLD §AutoBook	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-22	Booking state machine	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-23	Modify booking	HLD §Booking	F3: Battery-Aware Booking and Auto-Booking

Requirement ID	SRS description	SRS architecture reference	Owning module / contract
REQ-F3-24	Cancel booking	HLD \$Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-25	No-show handling	HLD \$Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-26	Auto-rebooking	HLD \$Booking	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-27	Check-in	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-28	Remote start	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-29	Live session monitor	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-30	Stop session	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-31	Start failure handling	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-32	Idle fee	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F3-33	Booking notifications	HLD \$Session	F3: Battery-Aware Booking and Auto-Booking
REQ-F4-01	Payment methods	HLD \$Payments	F4: Payments, Billing and Usage Analytics
REQ-F4-02	Tokenised payment methods	HLD \$Payments	F4: Payments, Billing and Usage Analytics
REQ-F4-03	Remove payment method	HLD \$Payments	F4: Payments, Billing and Usage Analytics
REQ-F4-04	Reservation fee	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-05	Final bill computation	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-06	Tariff snapshot	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-07	Settle balance	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-08	Refunds	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-09	Payment records	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-10	Idempotent retry	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-11	Webhook verification	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-12	Server-side verification	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-13	Daily reconciliation	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-14	Invoice generation	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-15	Payment history	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-16	Tariff management	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-17	Operator earnings	HLD \$Billing	F4: Payments, Billing and Usage Analytics
REQ-F4-18	Driver usage summary	HLD \$Analytics	F4: Payments, Billing and Usage Analytics

Requirement ID	SRS description	SRS architecture reference	Owning module / contract
REQ-F4-19	Driver trend charts	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-20	Top stations and CO2	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-21	Operator analytics	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-22	Platform analytics	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-23	Date range filter	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-24	CSV export	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-25	Role-scoped analytics	HLD §Analytics	F4: Payments, Billing and Usage Analytics
REQ-F4-26	Analytics refresh	HLD §Analytics	F4: Payments, Billing and Usage Analytics

15. Known Trade-offs, Risks and Validation Gates

Architecture is not a claim of working code. Booking concurrency, payment reconciliation, stale status, operator scoping and gateway timeouts are the highest-risk integration boundaries. The complete end-to-end demonstration cannot be shown until all four module contracts, the simulator and sandbox are integrated. The proposed backlog defers settlement and analytics to Sprint 6; this should be reviewed against actual term dates and measured team capacity. Requirement and API routes must be signed off before implementation. SRS Appendix C lists illustrative .py paths inconsistent with the SRS-fixed FastAPI implementation style in places; treat them as placeholders, not evidence of Django.

16. References

ChargeSlot SRS v1.0, Team NoteWare, 5 September 2026, §§2–6 and Appendix C; PES University UE24CS341A Jackfruit Software Engineering Project Guidelines 2026, Phase-3 template and final report format.